

LANCASTER COUNTY AP, SC, USA

WMO: 720607

Lat: 34.717N

Lon: 80.850W

Elev: 486

StdP: 14.44

Time Zone: -5.00 (NAE)

Period: 09-19

WBAN: 00199

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.5	23.2	0.4	5.7	29.7	6.7	7.9	32.8	19.4	53.3	16.8	52.0	1.2	50	0.369

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.9	95.0	75.1	92.9	74.6	90.9	74.6	79.7	86.7	78.5	85.7	77.7	84.8	5.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.5	145.6	80.9	77.0	143.3	80.5	75.4	135.5	79.2	43.5	87.7	42.5	85.9	41.4	85.1	84.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.1	12.8	11.0	12.4	98.2	5.1	3.0	8.7	100.4	5.8	102.2	3.0	103.9	-0.7	106.1		
			11.7	81.8	4.8	1.7	8.2	83.1	5.4	84.1	2.7	85.1	-0.7	86.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	61.6	41.5	46.0	51.8	61.4	69.9	77.2	80.0	78.5	73.4	62.1	50.6	46.1
	DBStd	15.43	10.04	9.86	9.56	7.53	6.35	4.22	3.55	3.73	5.54	7.81	7.82	9.43
	HDD50	870	302	181	96	7	0	0	0	0	5	84	195	
	HDD65	3056	730	535	420	156	34	0	0	0	149	434	591	
	CDD50	5111	37	69	151	349	617	816	931	883	703	379	103	73
	CDD65	1823	0	4	10	47	186	367	466	418	260	58	3	4
	CDH74	16596	2	16	112	517	1571	3502	4481	3693	2196	489	14	3
	CDH80	7208	0	0	11	111	533	1632	2207	1674	918	122	0	0
Wind	WSAvg	3.6	3.6	4.4	4.5	4.7	3.7	3.4	3.2	2.7	3.0	3.5	3.2	3.3
Precipitation	PrecAvg	41.5	3.7	3.1	3.9	3.1	2.5	4.4	4.1	4.4	3.4	3.2	2.9	3.4
	PrecMax	50.9	7.5	5.4	8.6	6.5	5.2	8.5	9.2	8.1	9.1	12.0	7.1	7.2
	PrecMin	30.3	1.0	1.6	0.8	0.9	0.9	0.9	1.6	0.8	0.0	0.0	0.4	1.3
	PrecStd	6.5	1.7	1.3	1.9	1.6	1.2	2.0	1.9	2.2	2.2	2.5	1.7	1.5
	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.6	76.8	81.5	86.0	92.9	97.4	99.2	95.2	93.4	89.6	75.4
MCWB			62.0	63.8	64.0	66.4	70.6	74.7	75.5	75.5	72.5	73.5	64.0	68.2
2%		DB	66.0	72.3	77.2	82.4	89.7	93.5	95.2	93.0	91.1	82.3	72.3	68.5
		MCWB	58.9	60.5	60.8	65.8	70.8	74.4	75.7	75.3	72.0	69.5	61.8	64.3
5%		DB	63.1	67.7	73.0	81.0	85.7	91.2	93.0	90.9	89.8	80.5	68.2	65.8
		MCWB	57.3	58.5	59.6	65.3	69.9	74.8	75.6	75.4	71.8	67.8	61.1	63.0
10%		DB	58.6	63.5	68.3	76.6	82.2	89.8	90.7	89.8	85.8	75.3	64.5	63.1
		MCWB	53.3	56.4	57.2	62.9	69.5	74.4	75.5	75.1	71.6	66.9	58.4	59.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.0	67.0	67.6	73.6	76.5	79.4	82.0	81.5	78.7	76.9	69.0	71.7
		MCDB	69.0	71.0	72.9	77.8	82.6	87.8	87.2	85.2	84.1	79.9	71.3	72.2
	2%	WB	63.0	64.2	65.5	71.2	74.8	78.3	79.8	79.9	76.9	74.7	66.3	66.4
		MCDB	64.3	67.8	73.3	76.3	80.2	86.8	87.6	85.8	82.2	78.7	68.8	67.7
	5%	WB	58.8	61.8	63.4	68.8	73.3	77.3	78.5	78.4	75.2	72.5	63.9	63.6
		MCDB	61.1	67.2	67.8	74.7	78.9	85.9	86.3	84.8	81.5	76.6	65.6	64.2
	10%	WB	54.2	57.7	60.8	66.5	72.2	76.2	77.7	77.4	73.5	69.5	60.9	59.1
		MCDB	56.6	61.4	67.0	72.9	78.4	83.7	85.2	83.7	79.5	74.1	65.3	61.9
Mean Daily Temperature Range	5% DB	MDBR	21.2	22.2	23.5	25.0	21.7	21.8	20.9	20.0	21.3	23.2	23.6	19.9
		MCDBR	25.6	25.4	27.4	28.2	25.4	25.2	24.0	22.7	25.9	25.6	26.4	20.2
		MCWBR	18.1	15.8	14.1	12.7	10.5	9.4	7.7	8.5	9.1	12.7	16.7	14.3
	5% WB	MCDBR	20.1	20.4	21.6	20.8	20.0	21.9	20.6	19.4	20.6	19.0	18.4	16.4
		MCWBR	17.4	15.4	14.1	11.8	9.8	9.6	8.7	8.8	9.2	11.3	15.0	14.2
Clear-Sky Solar Irradiance	taub	0.316	0.326	0.363	0.411	0.468	0.494	0.548	0.523	0.440	0.389	0.338	0.315	
	taud	2.500	2.470	2.397	2.253	2.137	2.105	1.978	2.036	2.251	2.368	2.471	2.537	
	Ebn at Noon	283	292	288	278	262	254	240	243	260	265	272	277	
	Edh at Noon	27	30	36	43	49	51	57	53	41	33	27	24	
All-Sky Solar Radiation	RadAvg	854	1057	1400	1753	1933	2040	1916	1749	1498	1237	941	739	
	RadStd	46	118	102	135	158	145	120	82	129	151	76	69	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (68 neighbors)	+0.97	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+306	+211	

Nomenclature: See separate page